



Sentiment Analysis & Generative AI Pipeline

An end-to-end framework for data processing,
Transformer fine-tuning, and LLM summarization.

TECHNICAL PROJECT REVIEW

01. Classification Model

Requirement : Transformer Fine-Tuning & Evaluation.



THE GRAY AREA PROBLEM

Ambiguous 3-Star Reviews

Standard mapping (1-2 neg, 3 neu, 4-5 pos) fails at the 3-star level. Sentiment in these cases is often mixed or context-dependent, leading to noise in the training data.

The Smart Filtering Approach

Isolated 3-star reviews and removed non-informative content (short/empty strings). This ensures the model only learns from high-quality linguistic candidates.

RoBERTa-Driven Inference

Used the 'twitter-roberta-base-sentiment-latest' model to compute Softmax probabilities. This identified reviews that are **genuinely** neutral based on language, not just stars.

Logical Grounding

The resulting hf_sentiment column serves as a ground truth that understands nuances, far exceeding the reliability of simple star ratings.

INFERENCE LOGIC

We leverage **Softmax Mathematics** to determine the most likely sentiment class.

By using a pre-trained models like **DistilBERT** and **RoBERTa**, we inherit a deep understanding of syntax and semantics, allowing us to filter candidates that represent the most valuable insights for our business model.



| DATASET BALANCE: THE KEY TO INSIGHT

57

Total Examples

9

193

Positive

193

Neutral

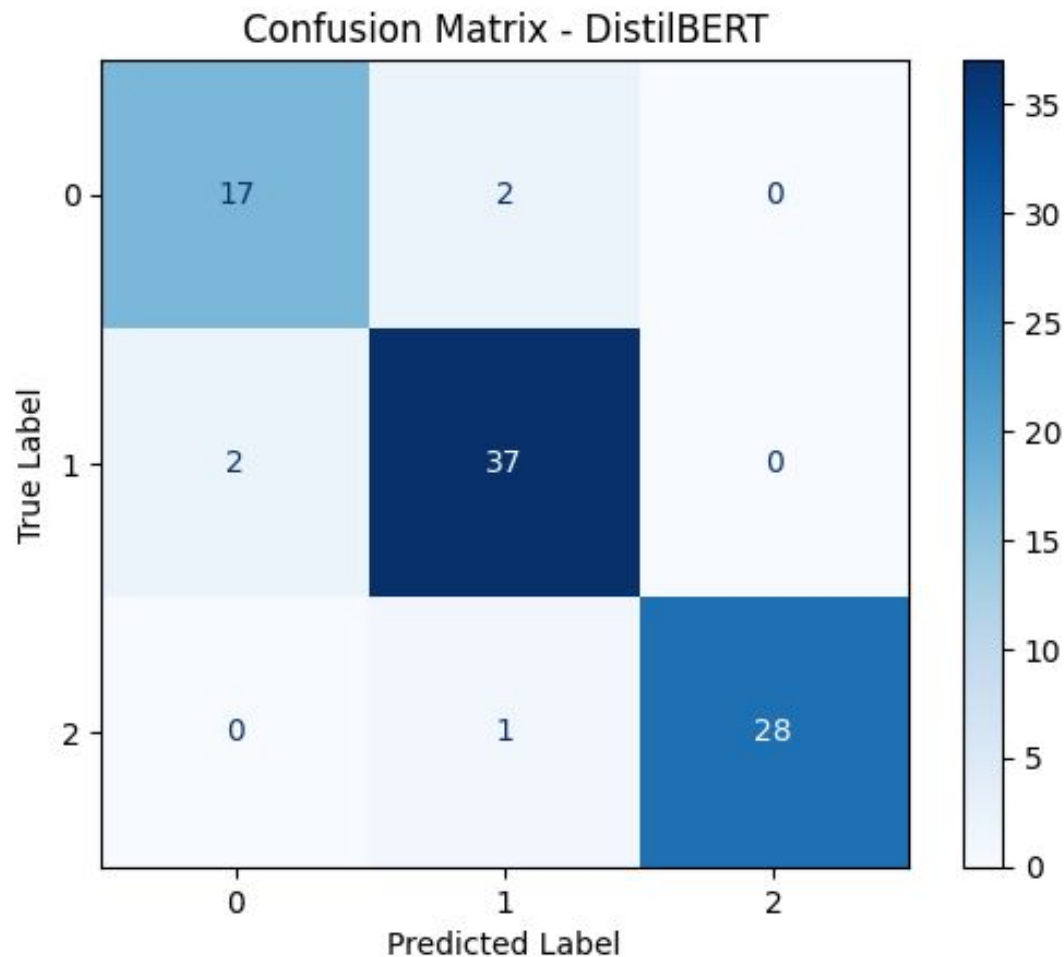
193

Negative

Why this works?

Fine-tuning a pre-trained model doesn't require millions of samples. A perfectly balanced dataset prevents class dominance and forces the model to master the neutral class—the most valuable for real insights.

DATASET BALANCE 01: DistilBERT



DistilBERT showed strong performance across all sentiment classes. The confusion matrix confirms high accuracy for negative, neutral, and positive reviews, with minimal misclassifications, mostly between similar negative and neutral texts. Positive reviews were rarely confused with negative ones. The model exhibits solid accuracy and balanced class performance.

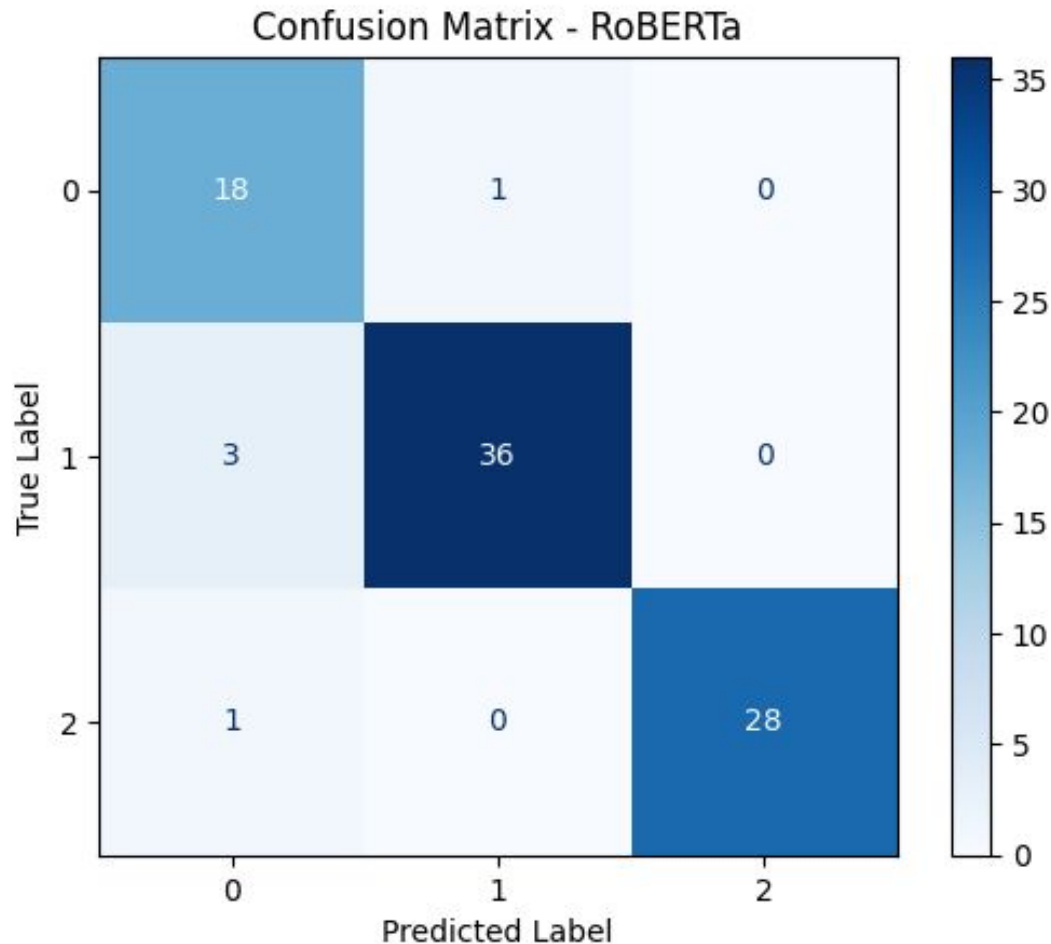
```
Classification report:
              precision    recall  f1-score   support

     0       0.8947      0.8947      0.8947        19
     1       0.9250      0.9487      0.9367        39
     2       1.0000      0.9655      0.9825        29

   accuracy              0.9425        87
  macro avg       0.9399      0.9363      0.9380        87
 weighted avg       0.9434      0.9425      0.9428        87
```

DATASET BALANCE 01:

RoBERTa ★



RoBERTa shows excellent performance, achieving 94% accuracy and a strong macro-F1 of 0.93. The confusion matrix confirms very few errors: negatives and positives are classified almost perfectly, and neutrals—typically the hardest class—are handled with high precision and recall. Overall, RoBERTa delivers a stable, reliable, and well-balanced sentiment classifier across all three classes.

Classification report:

	precision	recall	f1-score	support
0	0.8182	0.9474	0.8780	19
1	0.9730	0.9231	0.9474	39
2	1.0000	0.9655	0.9825	29
accuracy			0.9425	87
macro avg	0.9304	0.9453	0.9360	87
weighted avg	0.9482	0.9425	0.9439	87

| CLASSIFICATION STEP BY STEP

Data Preparation: Loading, cleaning, and combining review text.

Neutral Review Identification: Using a pre-trained Hugging Face (cardiffnlp/twitter-roberta-base-sentiment-latest) model to filter truly neutral 3-star reviews.

Dataset Balancing: Creating a balanced dataset with an equal number of negative, neutral, and positive reviews.

Model Training: Fine-tuning RoBERTa-base for sentiment classification (0=negative, 1=neutral, 2=positive).

Evaluation and Comparison: Analyzing the performance of both models using key metrics and confusion matrices.

Asset Saving: Persisting the balanced datasets and trained models for future project stages.

02. Clustering & Summarization

Requirements: From Vectors to Narrative Insights.



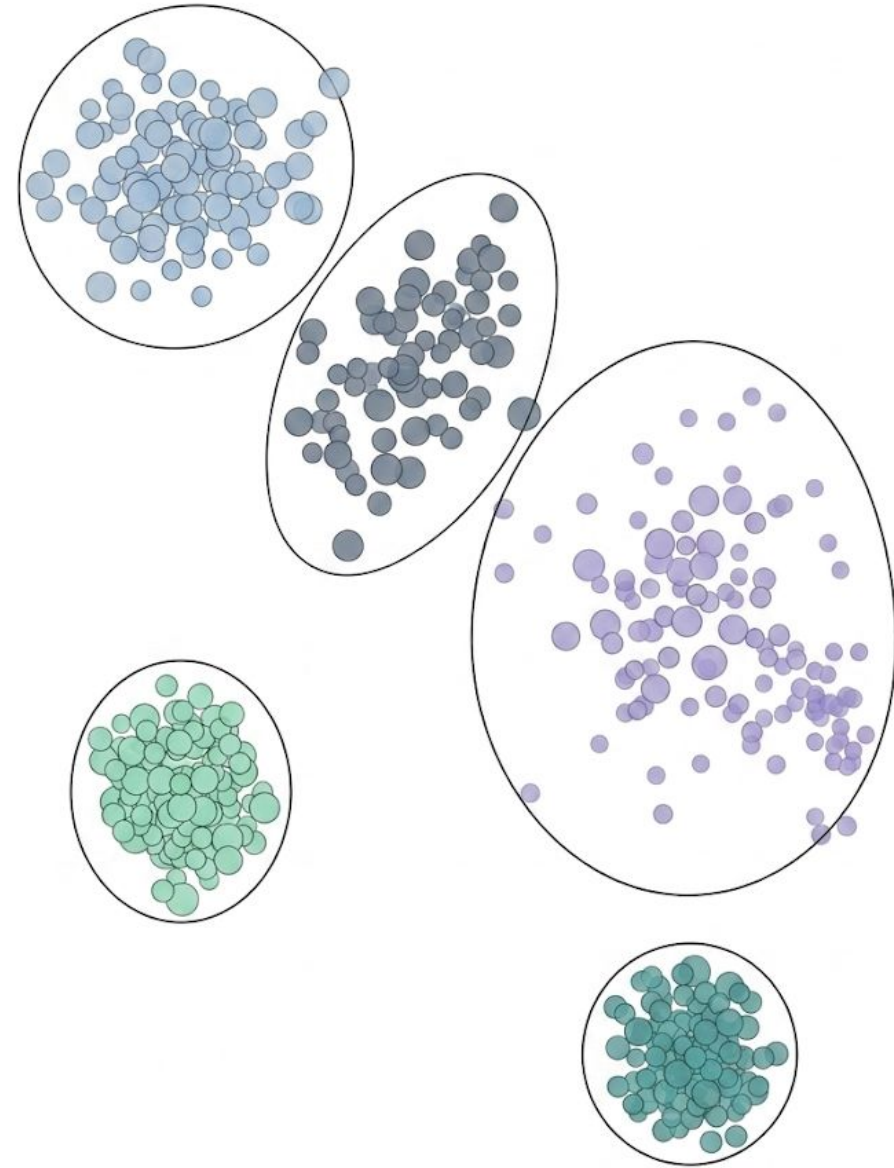
CLUSTERING

Data Preparation: Loading and balancing a dataset of Amazon reviews.

Embedding Generation: Using Sentence-BERT to convert review texts into numerical embeddings.

Dimensionality Reduction: Applying UMAP to reduce the complexity of these embeddings, making them easier to cluster and visualize.

Clustering: Employing both KMeans and HDBSCAN algorithms to group similar reviews into clusters based on their semantic content. The clusters are then visualized in 2D using **UMAP**.



WHY UMAP?

UMAP: Simplifying Complex Data - What is it?

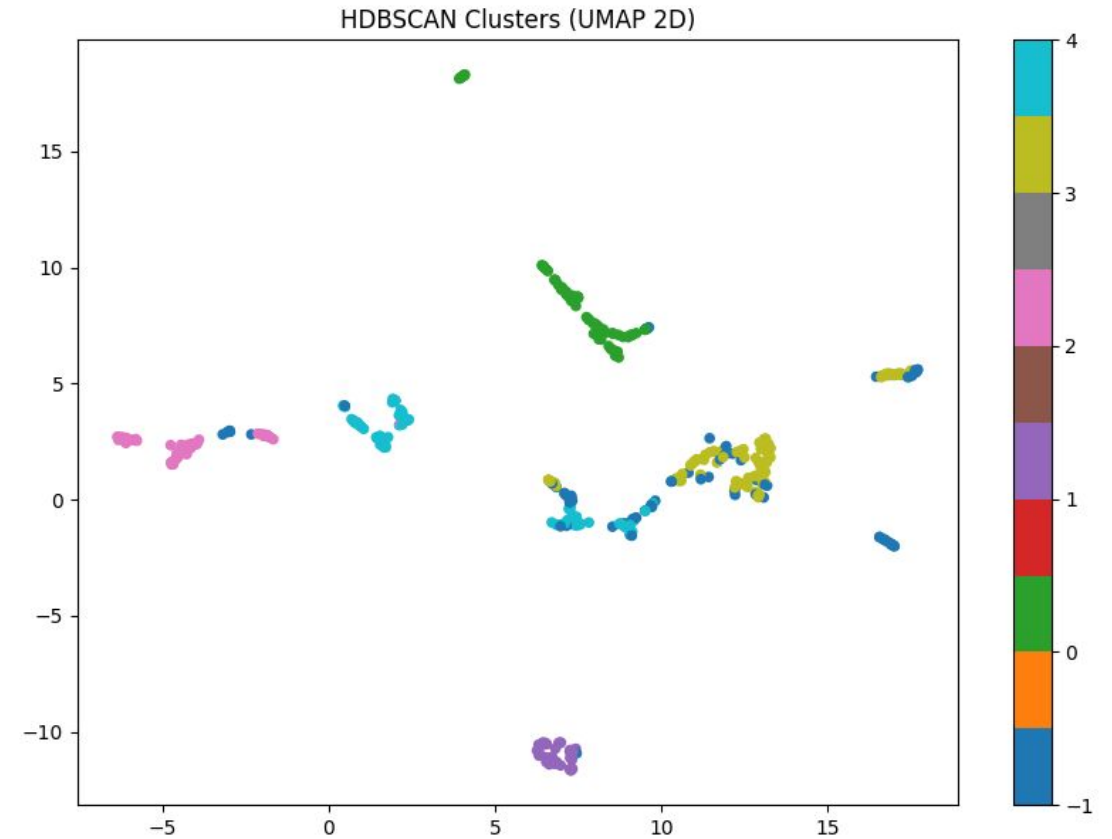
UMAP is a technique to simplify high-dimensional data (like our 384-dimension review embeddings) into fewer dimensions while preserving crucial relationships.

Why here?

Efficiency: Reduced 384D embeddings to 10D, making clustering faster and more effective.

Visualization: Further reduced to 2D to visually plot and understand the review clusters.

Achievement: Transformed complex review data into clear, actionable clusters and visual insights for better customer feedback analysis.



The UMAP plot shows review clusters in 2D. Points close together are reviews with similar meaning, while different colors represent different identified clusters.

| CLUSTERING - 2.0

Cluster Analysis and Naming: Each cluster was analyzed by extracting example reviews, average ratings, and main products using Pandas and `value_counts()`. Descriptive names were assigned to the clusters for easier interpretation.

Export for Summarization: JSON files and text blocks were generated for each cluster, ready to be processed by Large Language Models (LLMs).

Detailed Analysis (Product/Category): Cluster information was integrated into the original dataset using `Pandas.merge()`, and a detailed analysis by product and category was performed, identifying top/worst products and their key metrics.

LLM Prompt Generation: Python functions were created to structure specific and detailed prompts that enable LLMs to generate coherent and contextualized summaries for clusters, products, and categories.

GENERATIVE AI SUMMARIZATION



LLM Ingestion

An LLM reads aggregated texts from each cluster to generate cohesive narratives.



Thematic Reports

Output saved as structured JSON and individual text files per cluster.

AI REVIEW INTELLIGENCE PLATFORM

This app analyzes Amazon product reviews to uncover meaningful insights. It classifies each review as positive, neutral, or negative, identifies the best and worst-performing product categories, and highlights the top products within each category based on sentiment trends.

[ENTER HERE](#)

